

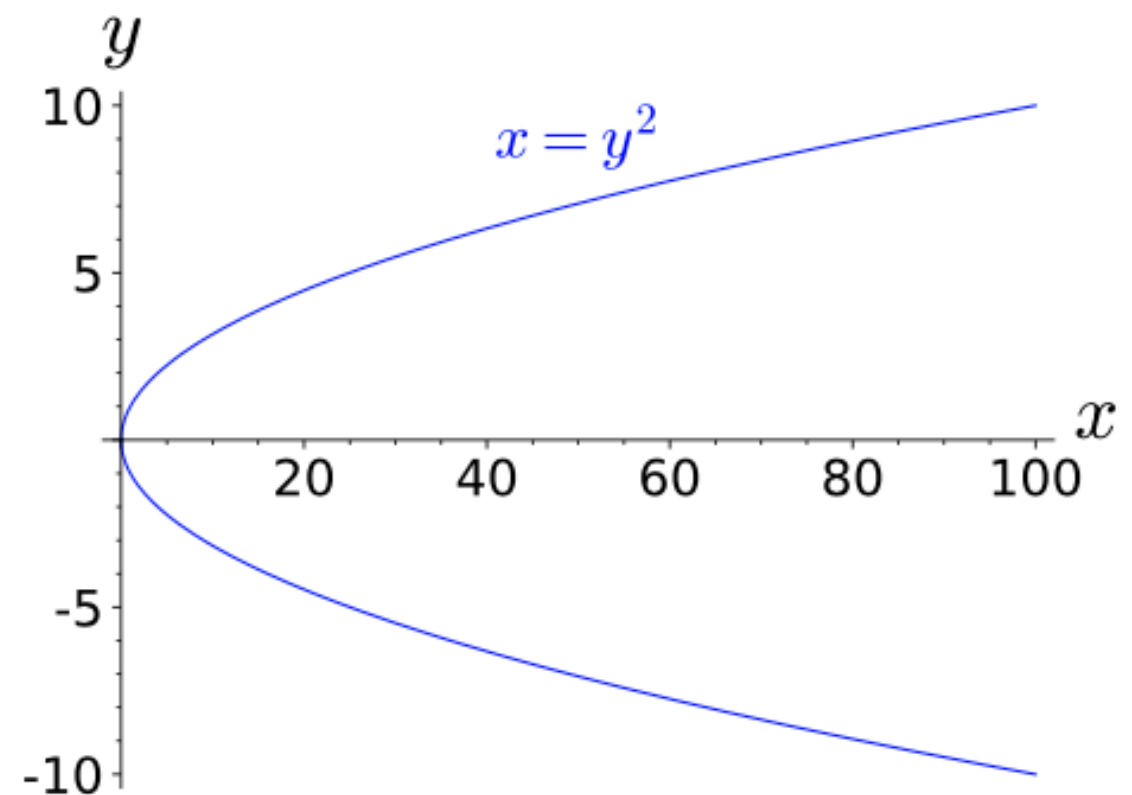
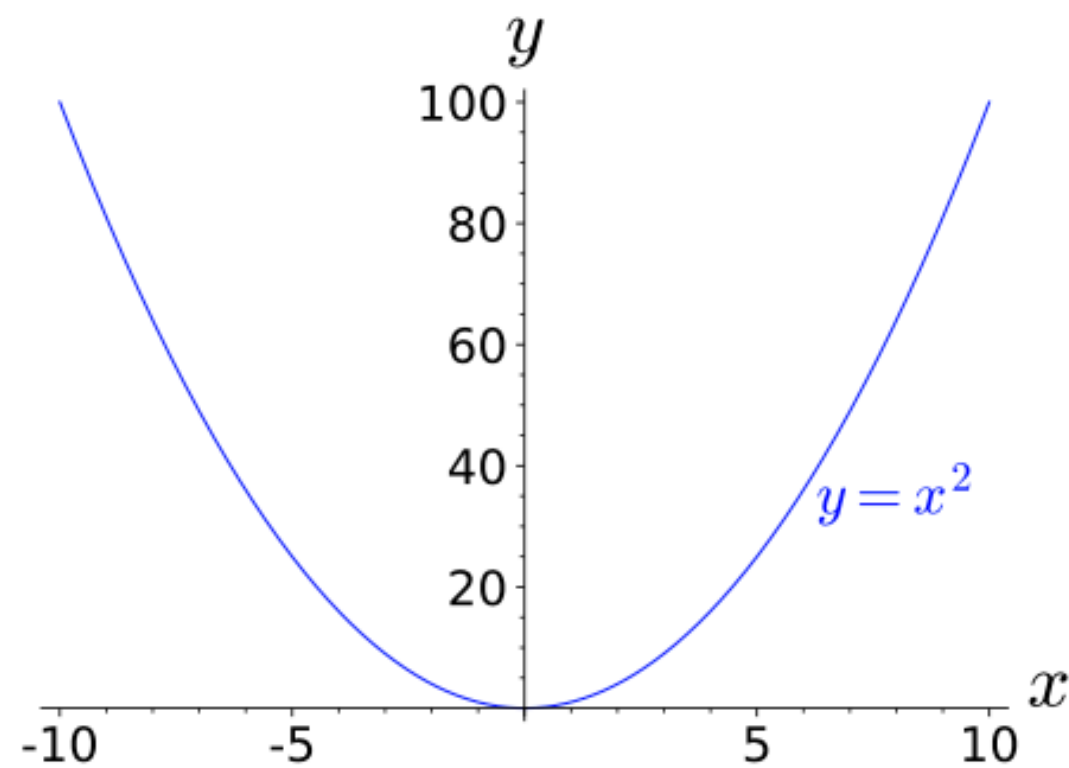
CSE113

Foundations of Computer Science

Relations

State University of New York, Korea

Functions vs. relations



	$y = x^2$	$y = \pm\sqrt{x}$
Function?	✓	✗
Relation?	✓	✓

Relation

Definition A **relation** on a set A is a subset $R \subseteq A \times A$. We often abbreviate the statement $(x, y) \in R$ as xRy . The statement $(x, y) \notin R$ is abbreviated as $x \not R y$.

$5 < 10$	$5 \leq 5$	$6 = \frac{30}{5}$	$5 \mid 80$	$7 > 4$	$x \neq y$	$8 \nmid 3$
$a \equiv b \pmod{n}$	$6 \in \mathbb{Z}$	$X \subseteq Y$	$\pi \approx 3.14$	$0 \geq -1$	$\sqrt{2} \notin \mathbb{Z}$	$\mathbb{Z} \not\subseteq \mathbb{N}$

Example: Less than

Problem

- A relation $L : \mathbb{R} \rightarrow \mathbb{R}$ as follows.
For all real numbers x and y , $(x, y) \in L \Leftrightarrow x L y \Leftrightarrow x < y$.
Draw the graph of L as a subset of the Cartesian plane $\mathbb{R} \times \mathbb{R}$.

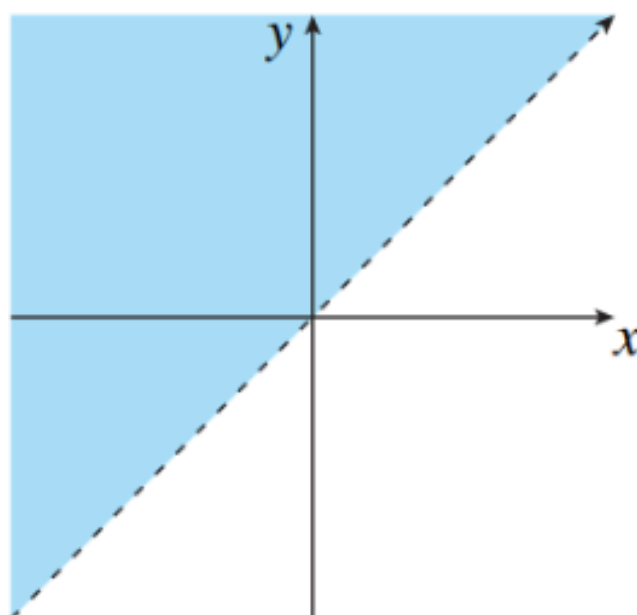
Example: Less than

Problem

- A relation $L : \mathbb{R} \rightarrow \mathbb{R}$ as follows.
For all real numbers x and y , $(x, y) \in L \Leftrightarrow x L y \Leftrightarrow x < y$.
Draw the graph of L as a subset of the Cartesian plane $\mathbb{R} \times \mathbb{R}$.

Solution

- $L = \{(-10.678, 30.23), (17.13, 45.98), (100/9, 200), \dots\}$
- Graph:



Example: Congruence modulo 2

Problem

- Define a relation $C : \mathbb{Z} \rightarrow \mathbb{Z}$ as follows.
For all $(m, n) \in \mathbb{Z} \times \mathbb{Z}$, $m C n \Leftrightarrow m - n$ is even.
- Prove that if n is any odd integer, then $n C 1$.

Example: Congruence modulo 2

Problem

- Define a relation $C : \mathbb{Z} \rightarrow \mathbb{Z}$ as follows.
For all $(m, n) \in \mathbb{Z} \times \mathbb{Z}$, $m C n \Leftrightarrow m - n$ is even.
- Prove that if n is any odd integer, then $n C 1$.

Solution

- $A = \{(2, 4), (56, 10), (-88, -64), \dots\}$
 $B = \{(7, 7), (57, 11), (-87, -63), \dots\}$
 $C = A \cup B$
- Proof. $(n, 1) \in C \Leftrightarrow n C 1 \Leftrightarrow n - 1$ is even
Suppose n is odd i.e., $n = 2k + 1$ for some integer k .
This implies that $n - 1 = 2k$ is even.

Example: Relation on a set

Problem

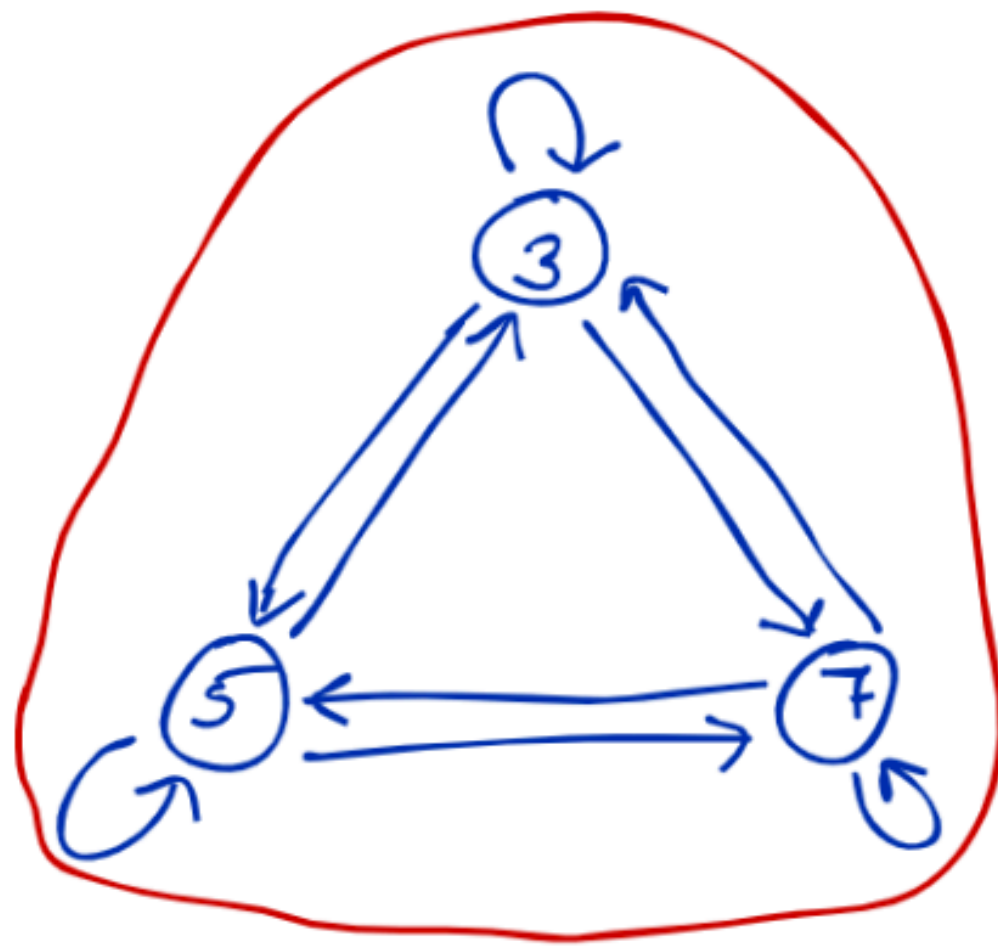
- Let $A = \{3, 4, 5, 6, 7, 8\}$. Define relation R on A as follows. For all $x, y \in A$, $x R y \Leftrightarrow 2 \mid (x - y)$. Draw the graph of R .

Example: Relation on a set

Problem

- Let $A = \{3, 4, 5, 6, 7, 8\}$. Define relation R on A as follows. For all $x, y \in A$, $x R y \Leftrightarrow 2 \mid (x - y)$. Draw the graph of R .

Solution



Quiz

The **congruence modulo 3** relation, T , is defined from $\mathbf{Z}$ to $\mathbf{Z}$ as follows: For all integers m and n ,

$$m T n \iff 3 \mid (m - n).$$

- a. Is $10 T 1$? Is $1 T 10$? Is $(2, 2) \in T$? Is $(8, 1) \in T$?
- b. List five integers n such that $n T 0$.
- c. List five integers n such that $n T 1$.
- d. List five integers n such that $n T 2$.

Inverse of a relation

Definition

- Let R be a relation from A to B .

Then **inverse relation** R^{-1} from B to A is:

$$R^{-1} = \{(y, x) \in B \times A \mid (x, y) \in R\}.$$

- For all $x \in A$ and $y \in B$,

$$(x, y) \in R \Leftrightarrow (y, x) \in R^{-1}.$$

Example: Inverse of a finite relation

Problem

- Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$.
Let $R : A$ to B . For all $(a, b) \in A \times B$, $a R b \Leftrightarrow a \mid b$
- Determine R and R^{-1} . Draw arrow diagrams for both.
Describe R^{-1} in words.

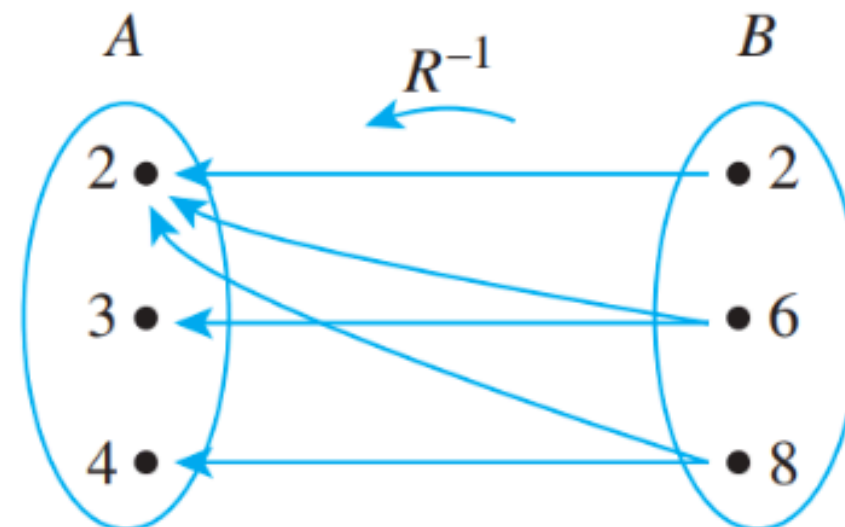
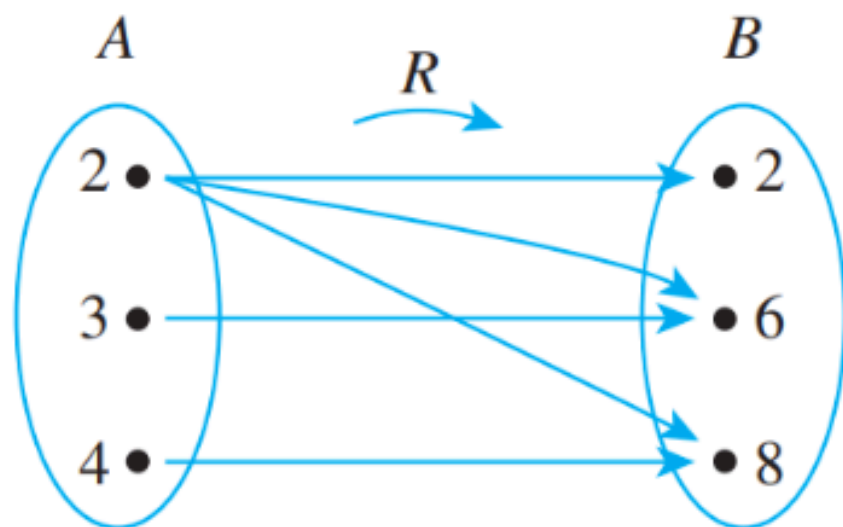
Example: Inverse of a finite relation

Problem

- Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$.
Let $R : A$ to B . For all $(a, b) \in A \times B$, $a R b \Leftrightarrow a \mid b$
- Determine R and R^{-1} . Draw arrow diagrams for both.
Describe R^{-1} in words.

Solution

- $R = \{(2, 2), (2, 6), (2, 8), (3, 6), (4, 8)\}$
 $R^{-1} = \{(2, 2), (6, 2), (8, 2), (6, 3), (8, 4)\}$
- For all $(b, a) \in B \times A$,
 $(b, a) \in R^{-1} \Leftrightarrow b$ is a multiple of a



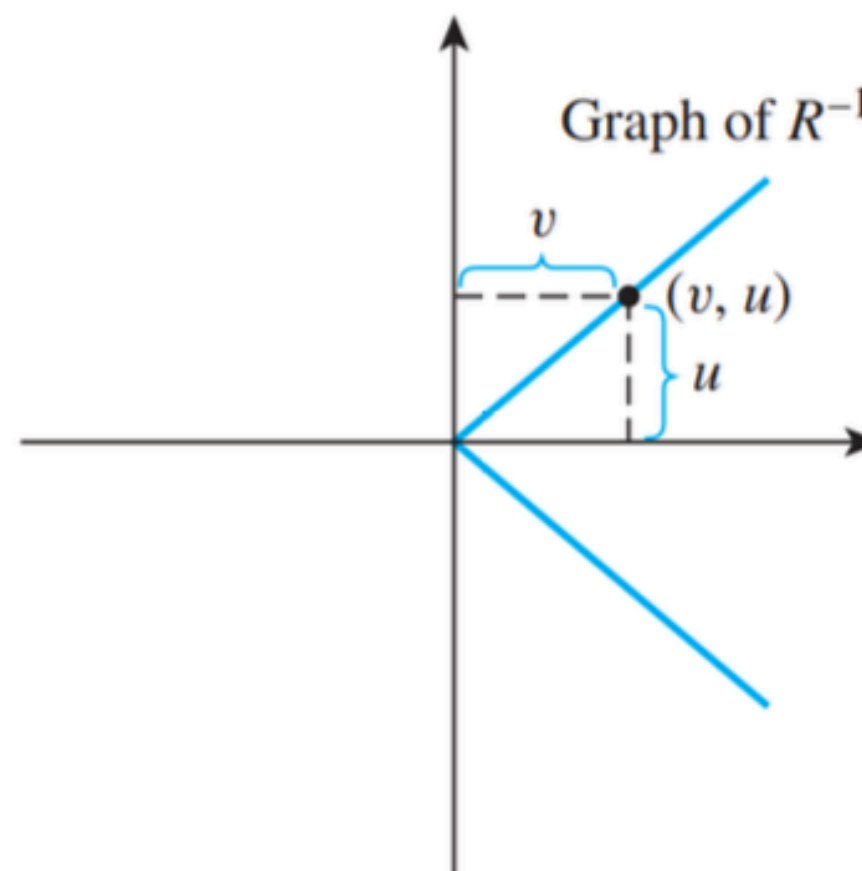
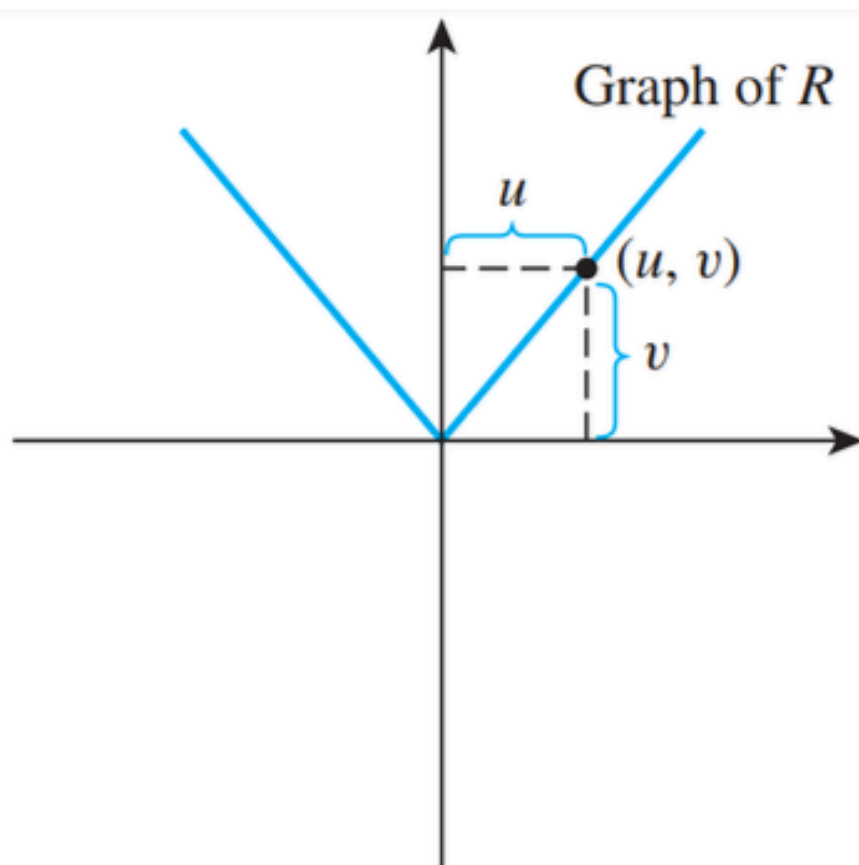
Example: Inverse of an infinite relation

Problem

- Define a relation R from $\mathbb{R}$ to $\mathbb{R}$ as follows:
For all $(u, v) \in \mathbb{R} \times \mathbb{R}$, $u R v \Leftrightarrow v = 2|u|$.
- Draw the graphs of R and R^{-1} in the Cartesian plane.
Is R^{-1} a function?

Solution

- R^{-1} is not a function. **Why?**



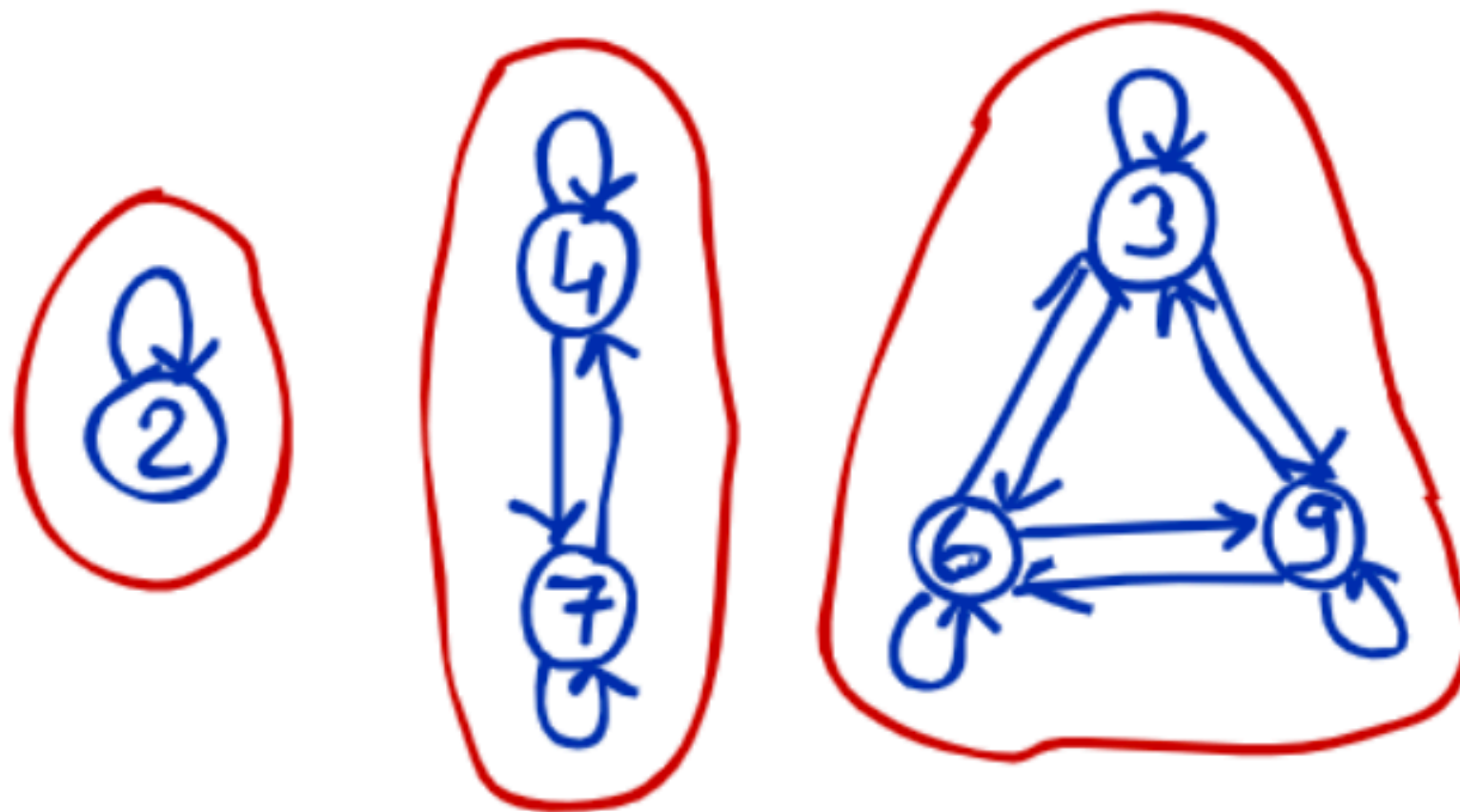
Equivalence Relation

Reflexivity, symmetry, and transitivity

Properties

- Set $A = \{2, 3, 4, 6, 7, 9\}$

Relation R on set A is: $\forall x, y \in A, x R y \Leftrightarrow 3 \mid (x - y)$



- Reflexivity.** $\forall x \in A, (x, x) \in R$.
- Symmetry.** $\forall x, y \in A$, if $(x, y) \in R$, then $(y, x) \in R$.
- Transitivity.**
 $\forall x, y, z \in A$, if $(x, y) \in R$ and $(y, z) \in R$, then $(x, z) \in R$.

Example

Problem

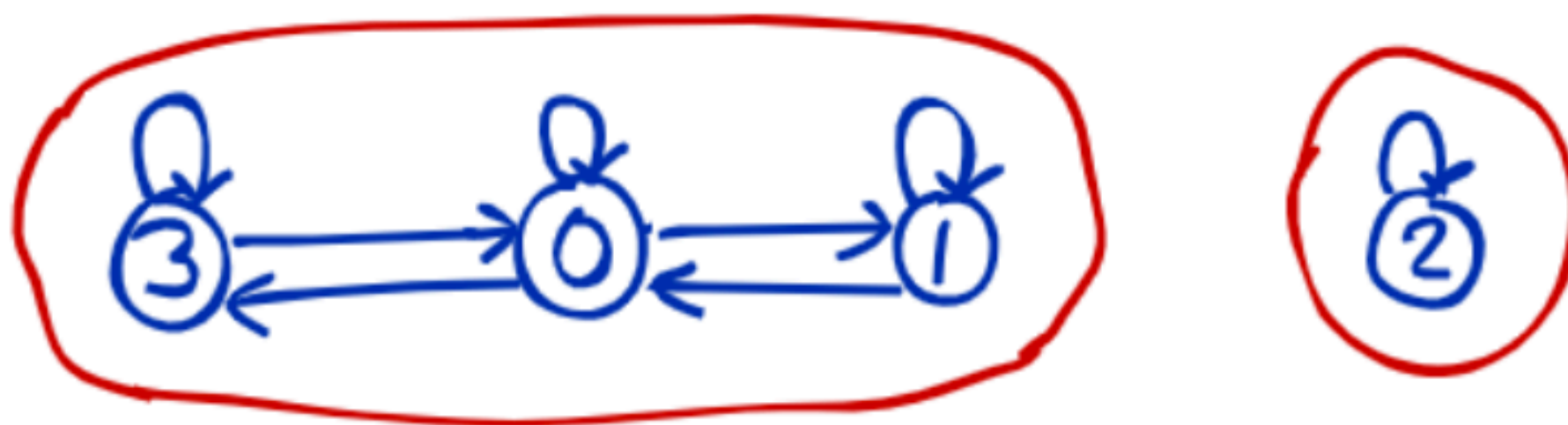
- $A = \{0, 1, 2, 3\}$.
 $R = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (2, 2), (3, 0), (3, 3)\}$.
Is R reflexive, symmetric, and transitive?

Example

Problem

- $A = \{0, 1, 2, 3\}$.
 $R = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (2, 2), (3, 0), (3, 3)\}$.
Is R reflexive, symmetric, and transitive?

Solution



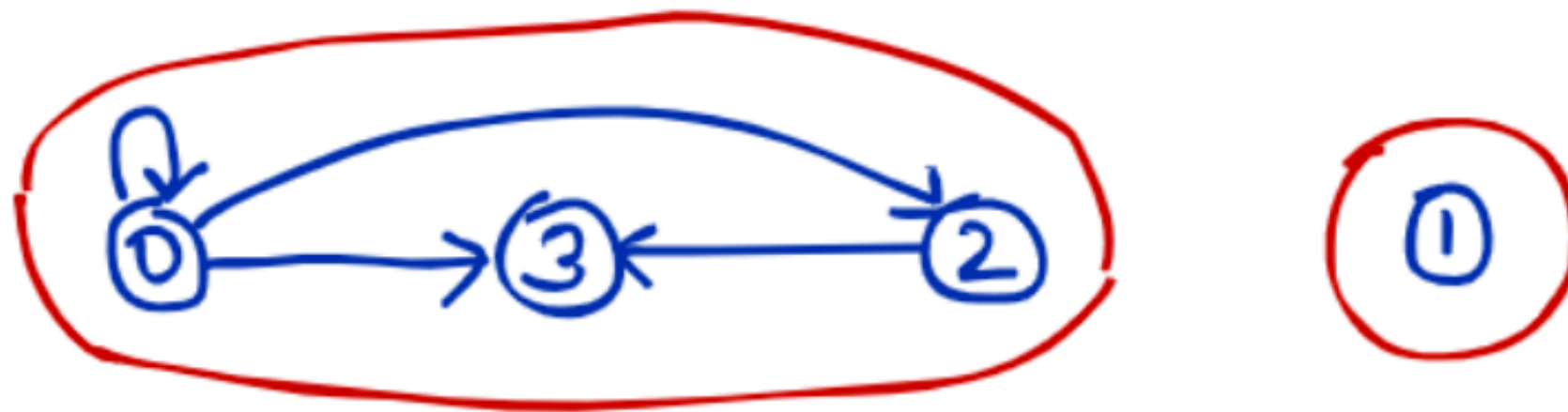
- **Reflexive.** $\forall x \in A, (x, x) \in R$.
- **Symmetric.** $\forall x, y \in A$, if $(x, y) \in R$, then $(y, x) \in R$.
- **Not transitive.** e.g.: $(1, 0), (0, 3) \in R$ but $(1, 3) \notin R$.
 $\exists x, y, z \in A$, if $(x, y) \in R$ and $(y, z) \in R$, then $(x, z) \notin R$.

Exercise 1

Problem

- $A = \{0, 1, 2, 3\}$. $R = \{(0, 0), (0, 2), (0, 3), (2, 3)\}$.
Is R reflexive, symmetric, and transitive?

Solution



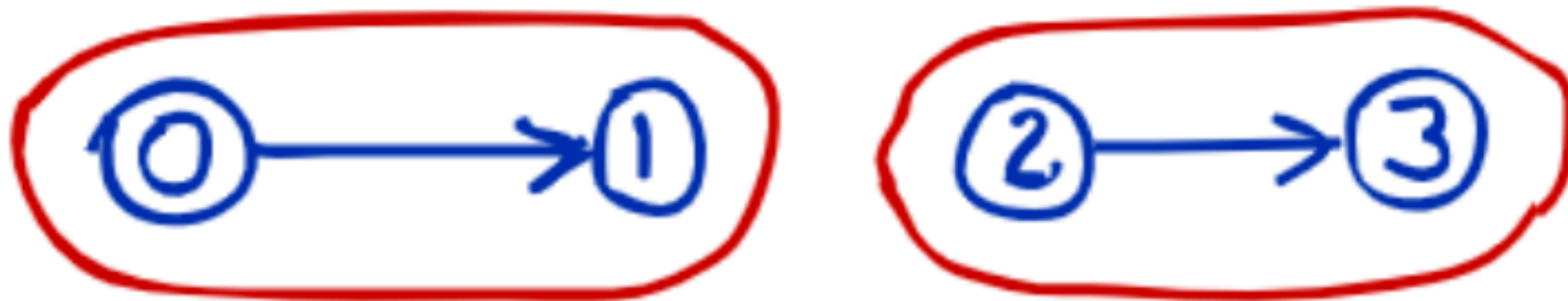
- **Not reflexive.** e.g.: $(1, 1) \notin R$. $\exists x \in A, (x, x) \notin R$.
- **Not symmetric.** e.g.: $(0, 3) \in R$ but $(3, 0) \notin R$.
 $\exists x, y \in A$, if $(x, y) \in R$, then $(y, x) \notin R$.
- **Transitive.**
 $\forall x, y, z \in A$, if $(x, y) \in R$ and $(y, z) \in R$, then $(x, z) \in R$.

Exercise 2

Problem

- $A = \{0, 1, 2, 3\}$. $R = \{(0, 1), (2, 3)\}$.
Is R reflexive, symmetric, and transitive?

Solution



- **Not reflexive.** e.g.: $(0, 0) \notin R$. $\exists x \in A, (x, x) \notin R$.
- **Not symmetric.** e.g.: $(0, 1) \in R$ but $(1, 0) \notin R$.
 $\exists x, y \in A$, if $(x, y) \in R$, then $(y, x) \notin R$.
- **Transitive.** Why?
 $\forall x, y, z \in A$, if $(x, y) \in R$ and $(y, z) \in R$, then $(x, z) \in R$.

Equivalence relation and equivalence class

Definition

- Relation R on set A is an **equivalence relation** iff R is reflexive, symmetric, and transitive.
- **Equivalence class** of element a , denoted by $[a]$, for an equivalence relation is defined as:
$$[a] = \{x \in A \mid (x, a) \in R\}.$$

Example: Less than

Problem

- Suppose R is a relation on $\mathbb{R}$ such that $x R y \Leftrightarrow x < y$.
Is R an equivalence relation?

Example: Less than

Problem

- Suppose R is a relation on $\mathbb{R}$ such that $x R y \Leftrightarrow x < y$.
Is R an equivalence relation?

Solution

- **Not reflexive.** e.g.: $0 \not< 0$. $\exists x \in \mathbb{R}, x \not< x$.
 - **Not symmetric.** e.g.: $0 < 1$ but $1 \not< 0$.
 $\exists x, y \in \mathbb{R}$, if $x < y$, then $y \not< x$.
 - **Transitive.** $\forall x, y, z \in \mathbb{R}$, if $x < y$ and $y < z$, then $x < z$.
- So, R is not an equivalence relation.

Example: Congruence modulo 3

Problem

- Suppose R is a relation on $\mathbb{Z}$ such that $m R n \Leftrightarrow 3 \mid (m - n)$.
Is R an equivalence relation?
- If yes, what is the equivalence class $[1]$, $[2]$ and $[3]$?

Example: Congruence modulo 3

Problem

- Suppose R is a relation on $\mathbb{Z}$ such that $m R n \Leftrightarrow 3 \mid (m - n)$. Is R an equivalence relation?

Solution

- **Reflexive.** $\forall m \in A, 3 \mid (m - m)$.
 - **Symmetric.** $\forall m, n \in A$, if $3 \mid (m - n)$, then $3 \mid (n - m)$.
 - **Transitive.**
 $\forall m, n, p \in A$, if $3 \mid (m - n)$ and $3 \mid (n - p)$, then $3 \mid (m - p)$.
- So, R is an equivalence relation.

Example: Congruence modulo 3

Solution

- **Equivalence classes.**

Three distinct equivalence classes are $[0]$, $[1]$, and $[2]$.

$$[0] = \{a \in \mathbb{Z} \mid a \equiv 0 \pmod{3}\} = \{0, \pm 3, \pm 6, \pm 9, \dots\}$$

$$[1] = \{a \in \mathbb{Z} \mid a \equiv 1 \pmod{3}\} = \{1, 1 \pm 3, 1 \pm 6, 1 \pm 9, \dots\}$$

$$[2] = \{a \in \mathbb{Z} \mid a \equiv 2 \pmod{3}\} = \{2, 2 \pm 3, 2 \pm 6, 2 \pm 9, \dots\}$$

Intuition.

$[0]$ = Set of integers when divided by 3 leave a remainder of 0.

$[1]$ = Set of integers when divided by 3 leave a remainder of 1.

$[2]$ = Set of integers when divided by 3 leave a remainder of 2.

Exercises

Exercise 1

- Let $A = \{3, 4, 5\}$ and $B = \{4, 5, 6\}$ and let R be the “less than” relation. That is, for all $(x, y) \in A \times B$,

$$x R y \iff x < y.$$

State explicitly which ordered pairs are in R and R^{-1} .

Exercise 2

- Determine if the following relations is reflective, symmetric, and transitive
 9. R is the “greater than or equal to” relation on the set of real numbers: For all $x, y \in \mathbf{R}$, $x R y \Leftrightarrow x \geq y$.
 10. C is the circle relation on the set of real numbers: For all $x, y \in \mathbf{R}$, $x C y \Leftrightarrow x^2 + y^2 = 1$.
 11. D is the relation defined on $\mathbf{R}$ as follows: For all $x, y \in \mathbf{R}$, $x D y \Leftrightarrow xy \geq 0$.
 12. E is the congruence modulo 2 relation on $\mathbf{Z}$: For all $m, n \in \mathbf{Z}$, $m E n \Leftrightarrow 2 \mid (m - n)$.
 13. F is the congruence modulo 5 relation on $\mathbf{Z}$: For all $m, n \in \mathbf{Z}$, $m F n \Leftrightarrow 5 \mid (m - n)$.
 14. O is the relation defined on $\mathbf{Z}$ as follows: For all $m, n \in \mathbf{Z}$, $m O n \Leftrightarrow m - n$ is odd.
 15. D is the “divides” relation on $\mathbf{Z}^+$: For all positive integers m and n , $m D n \Leftrightarrow m \mid n$.

Solution

- 9. R, Not S, T
 - 10. Not R, S, Not T
 - 11. R, S, Not T
 - 12. R, S, T
 - 13. R, S, T
 - 14. Not R, S, Not T
 - 15. R, Not S, T
9. R is the “greater than or equal to” relation on the set of real numbers: For all $x, y \in \mathbf{R}$, $x R y \Leftrightarrow x \geq y$.
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14. O is the relation defined on $\mathbf{Z}$ as follows: For all $m, n \in \mathbf{Z}$, $m O n \Leftrightarrow m - n$ is odd.
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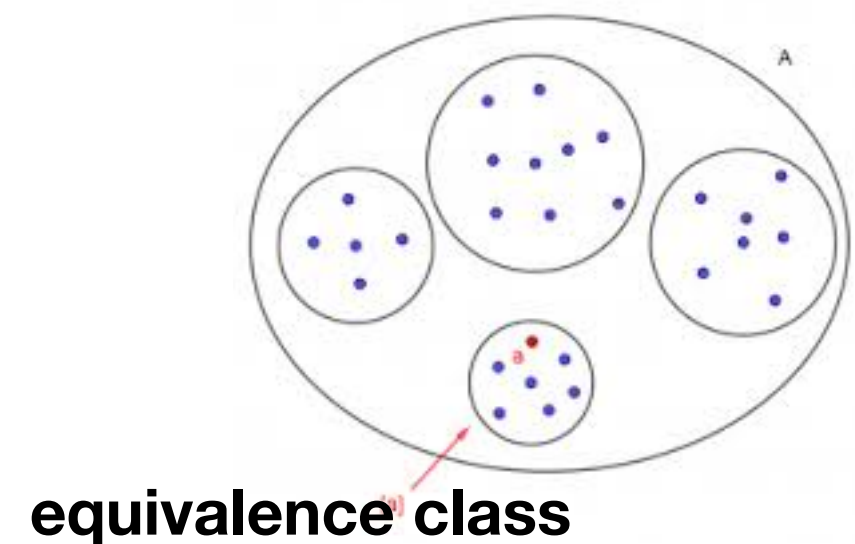
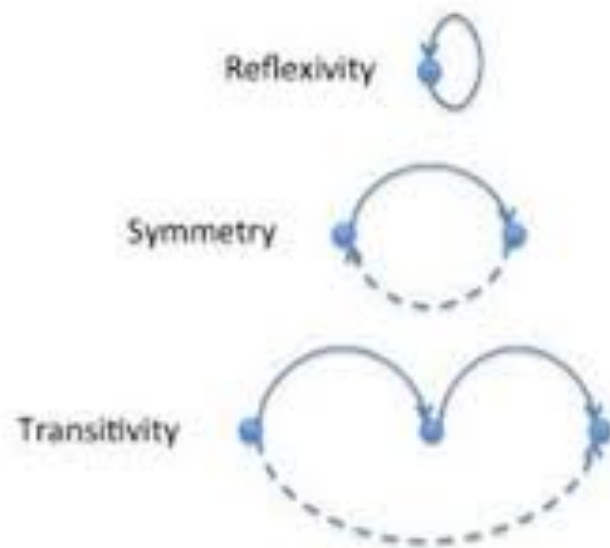
Exercise 3

2021 Final

Problem 11. [5 points]

Let A be the set of all people. Let R be the relation defined on A as follows: For persons p and q in A , we have $p R q \Leftrightarrow p$ has the same birthday as q .

Is R an equivalence relation? Prove your answer. If R is an equivalence relation, what are the distinct equivalence classes of the relation?



Solution

Problem 11. [5 points]

Let A be the set of all people. Let R be the relation defined on A as follows: For persons p and q in A , we have $p R q \Leftrightarrow p$ has the same birthday as q .

Is R an equivalence relation? Prove your answer. If R is an equivalence relation, what are the distinct equivalence classes of the relation?

- R is an equivalence relation, as it is
 - reflective ($p R p$),
 - symmetric $p R q \Leftrightarrow q R p$
 - transitive $p R q, q R r \Rightarrow p R r$
- Equivalence classes is the set of the sets of people having the same birthday.